

VYOMESH SHUKLA

shuklavyom12@gmail.com ,7505318821, [linkedin.com/in/vyomeshshukla/](https://www.linkedin.com/in/vyomeshshukla/)

Skills

- **Languages & Tools:** SQL (MySQL, MSSQL), C++, Java, JavaScript(Node.js), MongoDB, C,linux(Ubuntu,Debian) Python.
- **Web development skills:** HTML, CSS, JavaScript, React.js, Node.js, Express.js, MongoDB, Spring Boot, REST APIs, JWT authentication, Git/GitHub, and cloud deployment.
- **Soft Skills:** Cross-functional Collaboration, Creative Problem-Solving, Troubleshooting & Debugging, Adaptability & Fast Learning, Attention to Detail.
- **Core CS Skills:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, TCP/IP Networking Fundamentals, Databases

Work Experience

Data Analyst Intern at Codebasics

- Analyzed and solved business problems from hospitality data.
- Worked on uber data to streamline the booking process.

Projects (GitHub: <https://github.com/KNIGHT2000000>)

Distributed Payment Reliability & Retry Orchestrator | Java, Spring Boot, Kafka, PostgreSQL, Redis, Chaos Engineering

- Engineered an event-driven payment orchestrator across 5 microservices using Spring Boot, Kafka, and PostgreSQL, enforcing strict two-phase ledger updates and idempotent request execution via Redis-backed distributed locks.
- Designed a context-aware retry policy engine combining dynamic error classification, circuit breaking, and jittered exponential backoffs, reducing downstream retry amplification by 68% during injected PSP outages while sustaining a 99.4% transaction resolution rate.
- Built automated dead-letter and reconciliation workflows to resolve ambiguous HTTP 504 and dropped-webhook states, preventing duplicate downstream operations and ledger drift.
- Injected synthetic network partitions, latency spikes, and 5xx cascades using Chaos Mesh, benchmarking P99 latency and recovery time against baseline naïve exponential backoff via OpenTelemetry and Prometheus.

Offline Mesh UPI Settlement Backend | Java, Spring Boot, Distributed Systems, Cryptography

- Engineered a fault-tolerant Spring Boot backend for offline UPI payments, processing encrypted mesh-routed transactions via eventual consistency and deferred 4G synchronization.
- Implemented cryptographic payload decryption, idempotent processing, and automated deduplication to ensure secure ledger settlement and eliminate double-spending risks.
- Developed a software-based mesh network simulator to orchestrate multi-hop node handoffs, packet broadcasting, and edge-to-cloud batch uploads without requiring physical Bluetooth hardware.

Client-Server Architecture Implementation in C (TCP/IP Protocols)

- Designed and implemented a robust client-server architecture using C language over TCP/IP protocols.
- Enabled reliable two-way communication by developing socket programming logic for data exchange between nodes.
- Optimized connection stability and latency handling for real-time responsiveness.
- Applied multi-threading techniques to enhance server-side efficiency and scalability.

ProcWatch | Python, Java

- Developed a high-performance system monitoring service that interfaces directly with the Linux /proc virtual filesystem to

extract real-time kernel telemetry.

- Engineered a "pure file I/O" architecture to bypass heavy shell commands (top, ps), reducing overhead and eliminating dependencies on subprocess or exec() calls.
- Designed the service as the collector half of a distributed system, providing standardized JSON snapshots for a Java-based dashboard client.
- Leveraged Python 3.12 features and auto-generated Swagger UI documentation to ensure robust API schema validation and interactive testing.

Certifications

- Deep learning certifications on machine learning by Andrew Ng (Stanford)
- complete web development bootcamp by Udemy
- foundation in data by google
- Introduction to cloud computing and networking by Microsoft
- Complete node.js bootcamp by Udemy
- Data Analytics in python by CodeBasics

Education

B.Tech :(Computer Science and engineering) Symbiosis institute of Technology 2024- 2028

CGPA : 8.31

